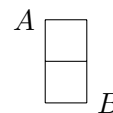


Name: _____

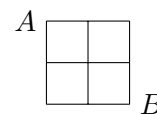
Date: _____

Hutch Mathcounts Handout: Pathfinding

- If Chong, at point A, wants to meet up with Pablo, at point B, by walking
1. on the segments of the diagram moving only to the right or downward, how many distinct paths are possible? _____



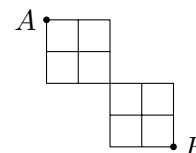
- Yash and Kifayat are looking for buried treasure that no one else can know about. They walk on the segments of the diagram from point A to point B moving only to the right or downward. How many distinct paths are possible? _____
- 2.



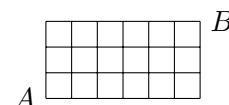
3. Austin really really really really really wants to know the answer to the following questions:

- a. How many different 4-letter words can we form by arranging the letters in RRDD? _____
- b. How does the answer to that relate to the answer of question 2? Why is this true?

- There's a Lubbock-wide Aaron meeting at 6:41 PM today. Aaron Cong and Aaron He will be attending, so they walk a perilous path from point A to point B by walking on the segments of the diagram and moving only to the right or downward. How many distinct paths are possible? _____
- 4.



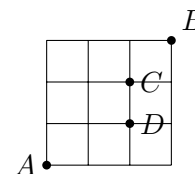
- If Rayirth wants to go to his favorite boba place, walking on the segments of the diagram to the right from point A to point B moving only to the right or upward, how many distinct paths are possible? _____
- 5.



Name: _____

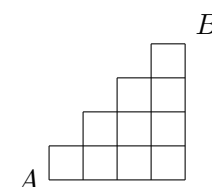
Date: _____

6. Chloe Hu, at point A, wants to get to Chloe Li, who's at the Lubbock-wide Chloe meeting at point B. She wants to avoid the Lubbock-wide Aaron meeting at point C, and she also has to take an emergency detour to the restroom at point D. She also only knows how to move right or up. How many paths can Chloe Hu take to get to the Lubbock-wide Chloe meeting? _____



7. A busy bee named Vincent lives in 3-dimensional space, starting at $(0, 0, 0)$. He can only increase a coordinate by 1 every minute. If he wants to get to Simon, at $(2, 4, 6)$, and he wants to take exactly 12 minutes getting there, how many distinct paths can he take? _____

8. Aiden thinks of a really good 6-7 joke and wants to tell it to Bryan, who's at the top right, so he walks on the segments of the diagram from point A to the top right, only moving only to the right or upward. How many distinct paths are possible? _____



9. Ainsley starts at point A and wants to jog to Yen, at point B, and every minute, she can either move any distance up or any distance right. For example, one such path is to go 2 units right, then 1 unit up, then 2 units up, then 1 unit right, then 1 unit up. In how many ways can Ainsley travel from A to B?
(Note that moving two units right in one move is now different from going 1 unit right and then 1 unit right again.) _____

